SOFTWARE REQUIREMENTS SPECIFICATION

QuantumGuard

Post Quantum Cryptography Scanner

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QuantumGuard	Draft

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for **QuantumGuard** — a comprehensive platform for organizational quantum readiness assessment, planning, and continuous improvement. QuantumGuard enables enterprises to systematically evaluate their vulnerability to quantum computing threats and transition to quantum-safe cryptography.

1.2 Scope

QuantumGuard encompasses:

- **Web-based Assessment Platform** — Interactive frontend providing Quick (12-question) and Comprehensive (120-question) assessments
- **Excel-based Toolkit** — Downloadable `QuantumGuard_Assessment_Toolkit.xlsx` with automated scoring, compliance mapping, and executive dashboards
- **Framework Documentation** — 8 core framework documents covering overview, maturity levels, scoring methodology, compliance mapping, and all 120 questions
- **Open Source Scanning Tools** — Three companion CLI tools (CryptoScan, TLS Analyzer, CryptoDeps) for automated cryptographic discovery
- **Templates Library** — Ready-to-use templates for executive reports, asset inventories, vendor questionnaires, and evidence collection
- **Implementation Guidance** — Step-by-step getting-started guide with phased implementation roadmap

1.3 Definitions, Acronyms, and Abbreviations

TERM	DEFINITION
QuantumGuard	Quantum Readiness Assurance Maturity Model — the renamed project (formerly QRAMM)
PQC	Post-Quantum Cryptography
CVI	Cryptographic Visibility & Inventory (Dimension 1)
SGRM	Strategic Governance & Risk Management (Dimension 2)
DPE	Data Protection Engineering (Dimension 3)
ITR	Implementation & Technical Readiness (Dimension 4)
ML-KEM	Module-Lattice-Based Key-Encapsulation Mechanism (FIPS 203)
ML-DSA	Module-Lattice-Based Digital Signature Standard (FIPS 204)
SLH-DSA	Stateless Hash-Based Digital Signature Standard (FIPS 205)
CNSA 2.0	NSA Commercial National Security Algorithm Suite
Y2Q	"Years to Quantum" — estimated time until cryptographically-relevant quantum computers
CBOM	Cryptographic Bill of Materials
SARIF	Static Analysis Results Interchange Format
CSNP	CyberSecurity NonProfit

1.4 References

REFERENCE	DESCRIPTION
NIST IR 8547	Transition to Post-Quantum Cryptography Standards
NIST SP 800-131A Rev. 2	Transitioning the Use of Cryptographic Algorithms
FIPS 203 / 204 / 205	NIST PQC Standards (ML-KEM, ML-DSA, SLH-DSA)
CMMC 2.0	Cybersecurity Maturity Model Certification
FedRAMP	Federal Risk and Authorization Management Program
ISO/IEC 27001:2022	Information Security Management
ETSI QSC	European Quantum-Safe Cryptography Standards
NIST CSF	NIST Cybersecurity Framework
NSM-10	National Security Memorandum — Quantum Readiness
M-23-02	OMB Memorandum on Quantum Computing Risks

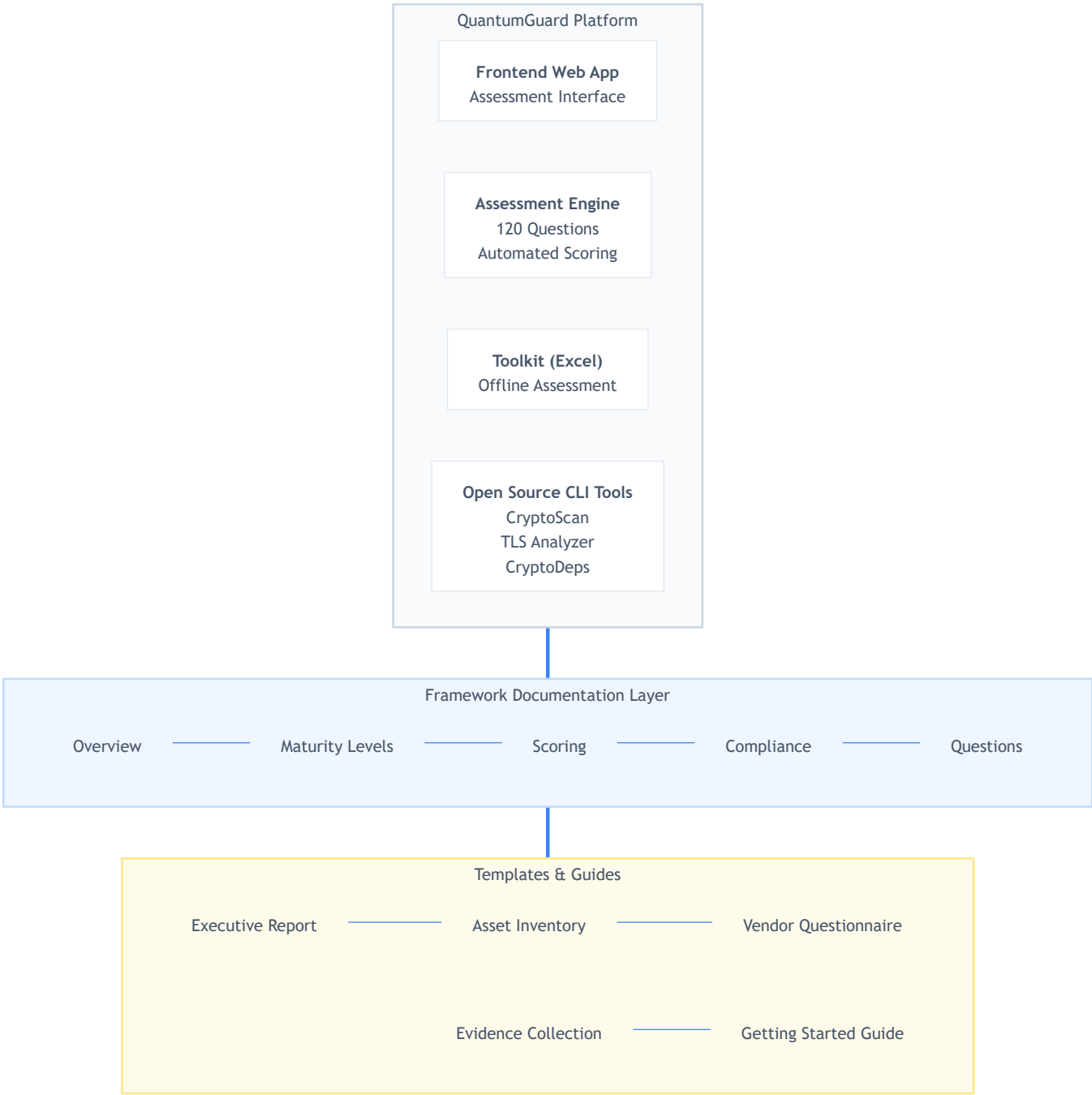
1.5 Document Overview

This SRS follows a modified IEEE 830 / ISO/IEC/IEEE 29148 structure. It covers both the existing system (framework, toolkit, website, companion tools) and defines requirements for their continued development under the QuantumGuard brand.

2. Overall Description

2.1 Product Perspective

QuantumGuard is an **open-source cybersecurity maturity framework** developed by CSNP. It is a standalone system composed of:



2.2 Product Functions

QuantumGuard performs the following high-level functions:

#	FUNCTION	DESCRIPTION
F1	**Quantum Readiness Assessment**	Evaluate organizational maturity across 4 dimensions, 12 practices, 120 questions
F2	**Automated Scoring**	Calculate practice, dimension, and overall scores with weighted adjustments
F3	**Maturity Level Classification**	Map scores to 5-level maturity model (Basic → Optimizing)
F4	**Compliance Mapping**	Map assessment results to 8 major regulatory frameworks
F5	**Gap Analysis**	Identify weaknesses and prioritize improvement areas
F6	**Executive Reporting**	Generate board-ready dashboards, charts, and recommendations
F7	**Cryptographic Discovery**	Scan codebases for quantum-vulnerable algorithms (CryptoScan)
F8	**TLS/SSL Analysis**	Evaluate cipher suites and certificates (TLS Analyzer)
F9	**Dependency Analysis**	Identify quantum-vulnerable algorithms in supply chain (CryptoDeps)
F10	**Template Generation**	Provide ready-to-use documents for implementation processes

2.3 User Classes and Characteristics

USER CLASS	DESCRIPTION	PRIMARY INTERACTIONS
Executive Leadership (CISO, CTO, CRO)	Strategic decision-makers needing high-level visibility	Quick Assessment, Executive Dashboard, Compliance Reports
Security Professionals	Technical practitioners conducting assessments and implementations	Comprehensive Assessment, CLI Tools, Gap Analysis
Compliance Teams	Regulatory and compliance staff managing audit requirements	Compliance Mapping, Evidence Collection, Vendor Questionnaires
IT Operations	Infrastructure and operations teams managing deployments	Implementation Guide, Infrastructure Assessment, CLI Tools
Third-Party Vendors	External partners subject to quantum readiness requirements	Vendor Questionnaire, Self-Assessment
Community Contributors	Open-source contributors improving the framework	GitHub Repository, Documentation

2.4 Operating Environment

COMPONENT	ENVIRONMENT
Frontend Web App	Modern web browsers (Chrome, Firefox, Safari, Edge) — responsive design supporting desktop and mobile
Excel Toolkit	Microsoft Excel 2016+ or compatible spreadsheet software (LibreOffice Calc)
CLI Tools	Cross-platform (Windows, macOS, Linux) — Node.js / Python runtime
Documentation	GitHub-hosted Markdown, rendered on project website
Repository	GitHub

2.5 Design and Implementation Constraints

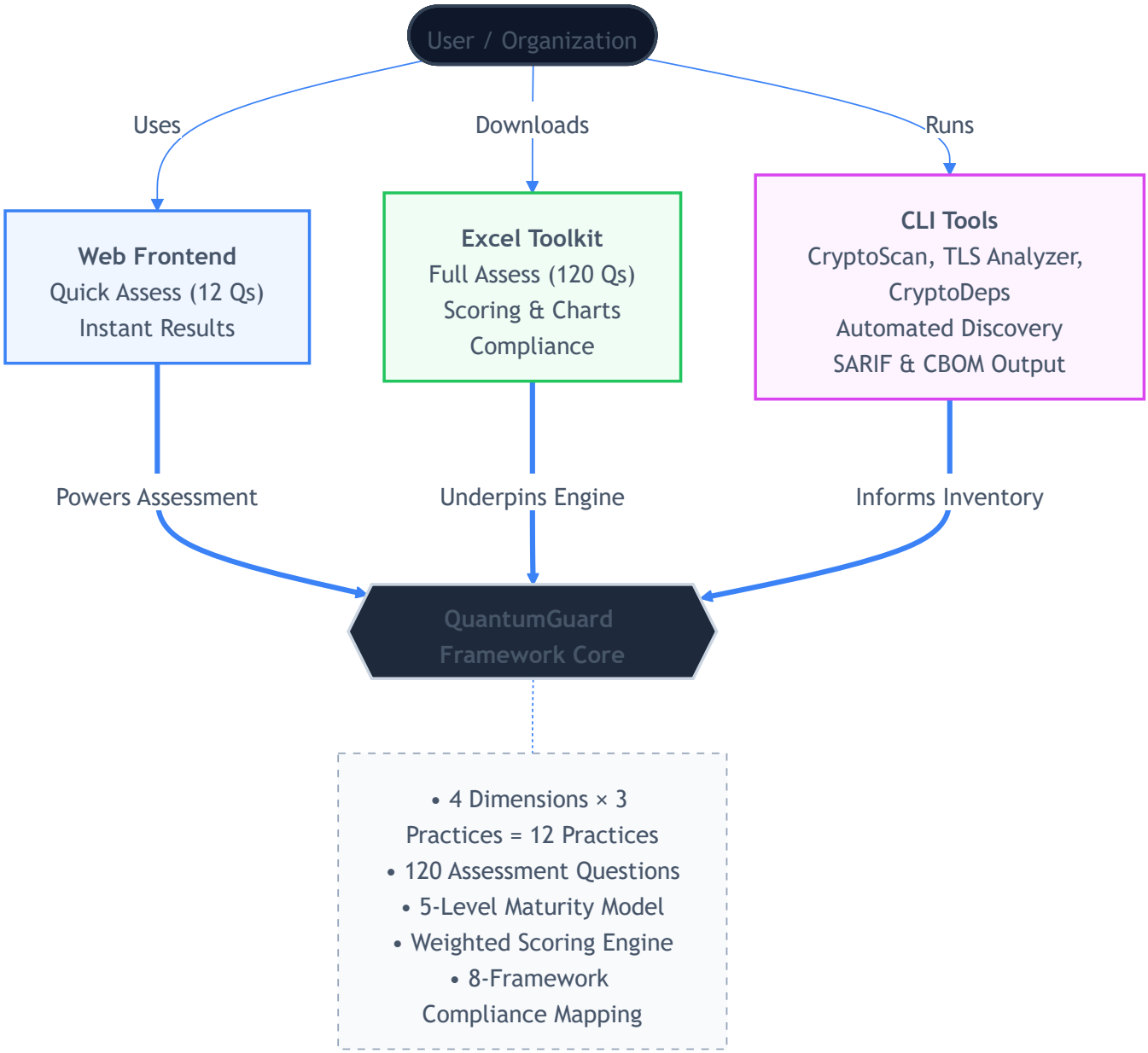
CONSTRAINT	DETAILS
License	MIT License — all components must remain open-source
Standards Alignment	Must remain aligned with active NIST PQC, CMMC, FedRAMP, FISMA, ISO 27001, ETSI QSC, CNSA 2.0, NSM-10 standards
Accessibility	Framework must remain accessible without requiring paid tools or proprietary software
Backward Compatibility	Assessment results from prior versions must remain interpretable
Offline Capability	Excel toolkit must function fully offline without internet connectivity

2.6 Assumptions and Dependencies

#	ASSUMPTION/DEPENDENCY
A1	Users have basic cybersecurity knowledge and organizational awareness
A2	Organizations have access to relevant stakeholders for comprehensive assessment
A3	NIST PQC standards continue to evolve and QuantumGuard must track updates
A4	The quantum threat timeline (5–15 years) remains the generally accepted industry estimate
A5	Excel-based toolkit relies on Microsoft Excel or compatible software with macro/formula support
D1	CryptoScan, TLS Analyzer, and CryptoDeps are maintained in separate GitHub repositories
D2	The QuantumGuard.org website is hosted and maintained independently

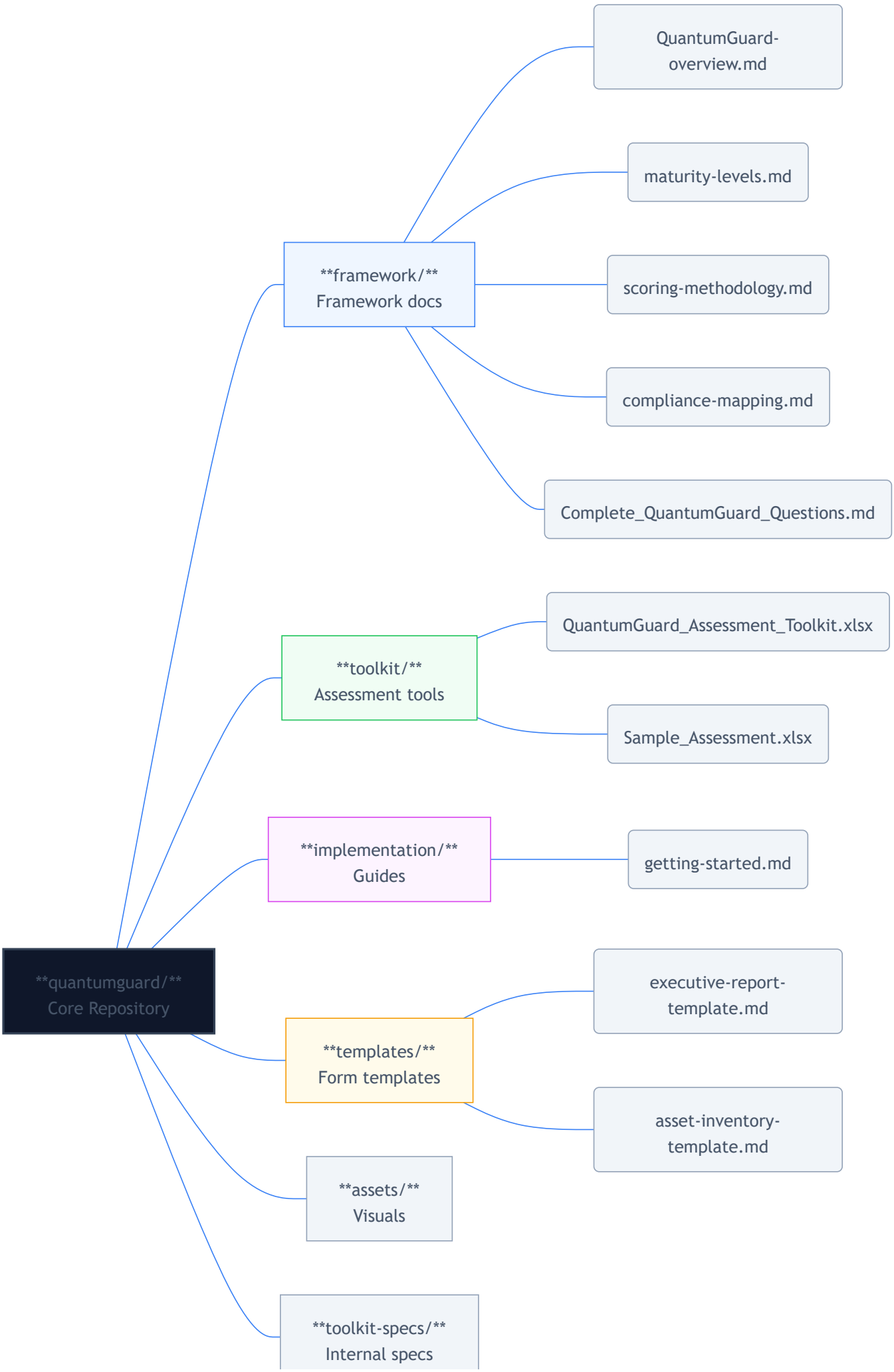
3. System Architecture

3.1 High-Level Architecture



3.2 Component Architecture

3.2.1 Repository Structure (Current)


quantumguard/
Core Repository
framework/
Framework docs
QuantumGuard-overview.md
maturity-levels.md
scoring-methodology.md
compliance-mapping.md
Complete_QuantumGuard_Questions.md
toolkit/
Assessment tools
QuantumGuard_Assessment_Toolkit.xlsx
Sample_Assessment.xlsx
implementation/
Guides
getting-started.md
templates/
Form templates
executive-report-template.md
asset-inventory-template.md
assets/
Visuals
toolkit-specs/
Internal specs

4. Functional Requirements

4.1 Assessment Engine

FR-001 — Quick Assessment

FIELD	VALUE
ID	FR-001
Priority	High
Description	The system shall provide a Quick Assessment mode with 12 questions (3 per dimension) that produces an instant baseline score and prioritized recommendations.
Input	12 multiple-choice responses (4 options each)
Output	Overall QuantumGuard score, maturity level, radar chart, per-dimension scores, priority recommendations
Acceptance Criteria	User completes in ≤10 minutes; results display immediately without page reload; PDF/print export available

FR-002 — Comprehensive Assessment

FIELD	VALUE
ID	FR-002
Priority	High
Description	The system shall provide a Comprehensive Assessment with 120 questions (30 per dimension, 10 per practice) organized in two streams: Stream A (Foundation, 60% weight) and Stream B (Advanced, 40% weight).
Input	120 multiple-choice responses (4 options each, scoring 1–4)
Output	Practice-level scores, dimension scores (MIN of practices), overall score, maturity level, gap analysis, compliance mapping
Acceptance Criteria	Supports save/resume; progress indicator shows completion %; all 12 practices covered

FR-003 — Scoring Calculation

FIELD	VALUE
ID	FR-003
Priority	High
Description	The system shall calculate scores using the defined methodology.
Formulas	`Practice Score = Σ(10 question scores) / 10`; `Dimension Score = MIN(Practice1, Practice2, Practice3)`; `Overall Score = (CVI + SGRM + DPE + ITR) / 4`
Maturity Thresholds	Level 1 (1.0–1.4), Level 2 (1.5–2.4), Level 3 (2.5–3.4), Level 4 (3.5–3.9), Level 5 (4.0)
Weighted Scoring	Organization Profile multiplier (0.8×–1.5×) adjusts raw scores based on industry, regulatory, scale, data sensitivity, tech complexity

FR-004 — Organization Profile Configuration

FIELD	VALUE
ID	FR-004
Priority	Medium
Description	The system shall capture organizational context to calculate a profile multiplier (0.8×–1.5×) based on: industry requirements, regulatory obligations, organizational scale, data sensitivity, and technology complexity.
Acceptance Criteria	Weighted scores can exceed 4.0 for high-risk profiles; N/A responses do not affect scoring

FR-005 — Gap Analysis

FIELD	VALUE
ID	FR-005
Priority	High
Description	The system shall identify and prioritize gaps at practice, stream, dimension, and question levels.
Gap Types	Practice-level gaps (below target), stream imbalances (Foundation vs Advanced), dimension weaknesses, critical deficiencies
Prioritization	By business impact, risk exposure, implementation complexity, timeline constraints

FR-006 — Save and Resume

FIELD	VALUE
ID	FR-006
Priority	Medium
Description	The system shall allow users to save assessment progress and resume later without data loss.
Acceptance Criteria	Progress persists across browser sessions; completion status accurately tracked

4.2 Compliance Mapping Module

FR-007 — Multi-Framework Compliance Mapping

FIELD	VALUE
ID	FR-007
Priority	High
Description	The system shall map each of the 120 assessment questions to controls in 8 compliance frameworks: NIST PQC, NSM-10, CNSA 2.0, ISO/IEC 27001:2022, ETSI QSC, CMMC 2.0, FedRAMP, and NIST CSF.
Mapping Strength	Direct (●●●), Strong (●●○), Partial (●○○), Supporting (○○○)
Compliance Score	`(Direct×1.0 + Strong×0.75 + Partial×0.5 + Supporting×0.25) / Total Requirements`

FR-008 — Compliance Coverage Report

FIELD	VALUE
ID	FR-008
Priority	Medium
Description	The system shall generate per-framework compliance coverage percentages and identify gaps requiring additional controls.
Output	Framework-specific coverage %, gap list, recommended actions

4.3 Reporting & Export

FR-009 — Executive Dashboard

FIELD	VALUE
ID	FR-009
Priority	High
Description	The system shall provide a board-ready executive dashboard showing overall score, maturity level, dimension summaries, top strengths, improvement areas, and dynamic recommendations.

FR-010 — Visual Analytics

FIELD	VALUE
ID	FR-010
Priority	High
Description	The system shall generate interactive charts: dimension scoring bar chart, maturity distribution donut chart, spider/radar chart comparing weighted scores against industry benchmarks.

FR-011 — Export Capabilities

FIELD	VALUE
ID	FR-011
Priority	Medium
Description	The system shall export assessment results in PDF, Excel, and print-friendly HTML formats.

4.4 Template Management

FR-012 — Template Library

FIELD	VALUE
ID	FR-012
Priority	Medium
Description	The system shall provide downloadable templates: Executive Report, Asset Inventory, Vendor Questionnaire, and Evidence Collection templates.

FR-013 — Template Customization

FIELD	VALUE
ID	FR-013
Priority	Low
Description	Templates shall auto-populate with organization name, assessment date, and scores where applicable.

4.5 Open Source Tool Integration

FR-014 — CryptoScan Integration

FIELD	VALUE
ID	FR-014
Priority	Medium
Description	The platform shall integrate with the CryptoScan tool to scan codebases for quantum-vulnerable algorithms, producing SARIF and CBOM output.
Supported Dimension	CVI (Dimension 1) — Cryptographic Visibility & Inventory

FR-015 — TLS Analyzer Integration

FIELD	VALUE
ID	FR-015
Priority	Medium
Description	The platform shall integrate with the TLS Analyzer to evaluate cipher suites and certificates with CNSA 2.0 compliance tracking.
Supported Dimension	DPE (Dimension 3) — Data Protection Engineering

FR-016 — CryptoDeps Integration

FIELD	VALUE
ID	FR-016
Priority	Medium
Description	The platform shall integrate with CryptoDeps to identify quantum-vulnerable algorithms in the software supply chain.
Supported Dimension	CVI (Dimension 1) — Cryptographic Visibility & Inventory

5. Frontend Requirements

5.1 Overview

The QuantumGuard frontend is a web application. It provides the primary user-facing interface for quantum readiness assessment, results visualization, and resource access.

5.2 Pages & User Flows

5.2.1 Landing Page / Home

FIELD	VALUE
ID	FE-001
Route	`/`
Purpose	Hero section introducing QuantumGuard; call-to-action for Quick Assessment; value proposition; threat timeline visualization; navigation to all sections
Key Elements	Hero banner with animated threat timeline; "Start Assessment" CTA button; feature highlights (4 dimensions); testimonials/badges (NIST Aligned, Enterprise Ready); open-source toolkit download links
Responsive	Desktop, tablet, mobile breakpoints

5.2.2 Quick Assessment Page

FIELD	VALUE
ID	FE-002
Route	`/assessment/quick`
Purpose	12-question rapid assessment covering all 4 dimensions
Key Elements	Progress stepper (12 steps); question card with 4 radio options per question; dimension indicator; "Next / Previous" navigation; submit button
UX Requirements	≤10 minute completion; immediate results on submit; no account required; smooth transitions between questions; mobile-friendly card layout

5.2.3 Comprehensive Assessment Page

FIELD	VALUE
ID	FE-003
Route	`/assessment/comprehensive`
Purpose	Full 120-question assessment organized by dimension → practice → question
Key Elements	Sidebar navigation tree (Dimensions > Practices > Questions); progress bar per dimension; save/resume functionality; question cards with evidence annotation field; completion status tracker
UX Requirements	Supports multi-session completion; auto-save every 30 seconds; keyboard-navigable; question filtering by answered/unanswered status

5.2.4 Results Dashboard

FIELD	VALUE
ID	FE-004
Route	`/results`
Purpose	Display assessment results with visual analytics and actionable recommendations
Key Elements	Overall QuantumGuard score (large numeral + maturity badge); 4-dimension score cards; radar/spider chart; maturity distribution donut chart; dimension bar chart; top 3 strengths + top 3 improvement areas; dynamic recommendations panel; compliance coverage summary; export buttons (PDF, Excel, Print)
Charts	Interactive (hover tooltips, click-to-drill-down); color-coded by maturity level (Red=Basic, Orange=Developing, Yellow=Established, Green=Advanced, Blue=Optimizing)

5.2.5 Compliance Mapping Page

FIELD	VALUE
ID	FE-005
Route	`/compliance`
Purpose	Show how QuantumGuard practices map to 8 compliance frameworks
Key Elements	Framework selector tabs (NIST PQC, NSM-10, CNSA 2.0, ISO 27001, ETSI QSC, CMMC, FedRAMP, NIST CSF); coverage percentage per framework; detailed mapping table (Question → Control → Mapping Strength); gap highlighter; evidence collection links

5.2.6 Open Source Tools Page

FIELD	VALUE
ID	FE-006
Route	`/tools`
Purpose	Showcase and guide users to the 3 companion CLI tools
Key Elements	Tool cards (CryptoScan, TLS Analyzer, CryptoDeps) with description, install command, link to GitHub repo; integration guide; output format documentation (SARIF, CBOM); dimension alignment indicator

5.2.7 Documentation / Framework Page

FIELD	VALUE
ID	FE-007
Route	`/docs`
Purpose	Serve framework documentation in a searchable, navigable format
Key Elements	Document sidebar (Overview, Maturity Levels, Scoring, Compliance, Questions); markdown rendering; search functionality; section anchoring; print-friendly styles

5.2.8 Templates & Downloads Page

FIELD	VALUE
ID	FE-008
Route	`/downloads`
Purpose	Provide downloadable resources
Key Elements	Download cards for: QuantumGuard Assessment Toolkit (.xlsx), Sample Assessment (.xlsx), Executive Report Template (.md), Asset Inventory Template (.md), Vendor Questionnaire Template (.md), Evidence Collection Template (.md); Getting Started Guide link

5.2.9 About / Leadership Page

FIELD	VALUE
ID	FE-009
Route	`/about`
Purpose	Information about CSNP, framework authors, and community
Key Elements	Mission statement; author profiles (Emily Fane, Abdel Fane); CSNP information; LinkedIn links; contact form/email; community resources (GitHub Discussions, Issues)

5.3 Frontend Component Specifications

5.3.1 Global Components

COMPONENT	DESCRIPTION
Navigation Bar	Sticky top nav with logo (QuantumGuard), links: Home, Assessment, Results, Compliance, Tools, Docs, Downloads, About; mobile hamburger menu
Footer	CSNP branding, license info, social links, copyright, sitemap links
Theme System	Dark mode (primary) with optional light mode toggle; color palette: quantum-themed (deep blues, purples, cyan accents)
Notification Toast	Success/error/info messages (save confirmation, export ready, etc.)
Loading States	Skeleton loaders for chart rendering; spinner for pdf generation

5.3.2 Assessment Components

COMPONENT	DESCRIPTION
QuestionCard	Displays question text, 4 radio-button options, optional evidence annotation textarea, dimension/practice badge
ProgressStepper	Visual step indicator showing current position in assessment flow; clickable steps for navigation
DimensionSidebar	Collapsible tree: 4 Dimensions → 3 Practices each → 10 Questions each; completion checkmarks
ScoreGauge	Circular gauge displaying score (1.0–4.0) with color-coded maturity ring
ComplianceBadge	Small badge showing framework name + coverage % with color coding

5.3.3 Visualization Components

COMPONENT	DESCRIPTION
RadarChart	4-axis radar chart showing dimension scores vs benchmarks; interactive with hover data
BarChart	Grouped bar chart for dimension comparison (raw vs weighted vs benchmark)
DonutChart	Maturity distribution showing % of practices at each level
HeatmapGrid	12-cell grid (4 dimensions × 3 practices) color-coded by maturity level
TimelineWidget	Interactive quantum threat timeline (2024–2033+) with organizational position marker

5.4 Frontend Design Requirements

5.4.1 Visual Design

REQUIREMENT	SPECIFICATION
Design Language	Modern, premium, security-focused aesthetic with glassmorphism and dynamic gradients
Color Palette	Primary: Deep Navy (#0A192F), Accent: Quantum Cyan (#00D4FF), Secondary: Violet (#7C3AED), Surface: Dark Glass (#1A1B2E), Success: Emerald (#10B981), Warning: Amber (#F59E0B), Danger: Rose (#F43F5E)
Typography	Primary: Inter (headings), Secondary: JetBrains Mono (scores/code), Body: system-ui stack
Iconography	Lucide Icons or Heroicons; quantum-themed custom icons for dimensions
Motion	Smooth page transitions (300ms ease); number counter animations for scores; chart entry animations; subtle hover micro-interactions
Spacing	8px grid system; consistent padding/margin scales

5.4.2 Responsive Breakpoints

BREAKPOINT	WIDTH	BEHAVIOR
Mobile	< 768px	Single column; hamburger nav; stacked cards; simplified charts
Tablet	768px – 1024px	Two-column layouts; collapsible sidebar; touch-friendly targets (44px min)
Desktop	> 1024px	Full layout with sidebar; multi-panel dashboard; expanded charts

5.4.3 Accessibility

REQUIREMENT	STANDARD
WCAG Compliance	WCAG 2.1 Level AA minimum
Keyboard Navigation	All interactive elements navigable via keyboard; visible focus indicators
Screen Readers	ARIA labels on all interactive elements; semantic HTML structure
Color Contrast	4.5:1 minimum contrast ratio for all text
Motion	Respect `prefers-reduced-motion` media query

5.5 Frontend State Management

STATE	SCOPE	PERSISTENCE
Assessment Responses	Session-scoped; 120 answer slots	LocalStorage + optional server-side save
Organization Profile	Session-scoped	LocalStorage
Calculated Scores	Derived from responses	Re-computed on change
Theme Preference	User preference	LocalStorage
Assessment Progress	Derived from response count	Computed

5.6 Frontend Performance Requirements

METRIC	TARGET
First Contentful Paint	< 1.5s
Largest Contentful Paint	< 2.5s
Time to Interactive	< 3.5s
Cumulative Layout Shift	< 0.1
Lighthouse Score	≥ 90 (Performance, Accessibility, SEO, Best Practices)
Bundle Size	< 500KB gzipped (initial load)

6. Non-Functional Requirements

6.1 Performance

ID	REQUIREMENT
NFR-001	Score calculations shall complete in < 100ms for the full 120-question assessment
NFR-002	Chart rendering shall complete in < 500ms after data is available
NFR-003	PDF report generation shall complete in < 5 seconds
NFR-004	Excel toolkit shall handle scoring for all 120 questions without macro performance degradation

6.2 Availability

ID	REQUIREMENT
NFR-005	Web application shall target 99.5% uptime
NFR-006	Excel toolkit shall function 100% offline
NFR-007	Framework documentation shall be available on GitHub with CDN-backed website

6.3 Scalability

ID	REQUIREMENT
NFR-008	Web application shall support 1,000+ concurrent users
NFR-009	Assessment data architecture shall support future expansion beyond 120 questions
NFR-010	Compliance mapping shall be extensible to additional frameworks without code changes

6.4 Usability

ID	REQUIREMENT
NFR-011	Quick Assessment shall be completable by a non-technical user in ≤10 minutes
NFR-012	Comprehensive Assessment shall be completable in 2–4 hours with stakeholder input
NFR-013	Results shall be understandable by executive-level audiences without technical training
NFR-014	All UI text shall use plain language; jargon shall include tooltip explanations

6.5 Maintainability

ID	REQUIREMENT
NFR-015	Assessment questions shall be data-driven (editable without code changes)
NFR-016	Compliance mappings shall be stored in structured data files (JSON/YAML)
NFR-017	All documentation shall follow consistent Markdown formatting
NFR-018	Codebase shall include inline documentation and a contribution guide

6.6 Portability

ID	REQUIREMENT
NFR-019	CLI tools shall run on Windows, macOS, and Linux
NFR-020	Web application shall function in all modern browsers (Chrome, Firefox, Safari, Edge)
NFR-021	Excel toolkit shall be compatible with Excel 2016+ and LibreOffice Calc 7.0+

6.7 Localization

ID	REQUIREMENT
NFR-022	Framework documentation and UI shall initially support English (US)
NFR-023	Architecture shall support future multi-language localization through resource files

7. Data Requirements

7.1 Assessment Data Model

```
Organization
├── Profile
│   ├── name: string
│   ├── industry: enum
│   ├── size: enum (Small, Medium, Large, Enterprise)
│   ├── regulatory_obligations: string[]
│   ├── data_sensitivity: enum (Low, Medium, High, Critical)
│   ├── technology_complexity: enum (Basic, Moderate, Complex)
│   └── profile_multiplier: float (0.8-1.5)
│
├── Assessment
│   ├── id: uuid
│   ├── type: enum (Quick, Comprehensive)
│   ├── started_at: datetime
│   ├── completed_at: datetime
│   ├── status: enum (InProgress, Completed)
│   │
│   ├── Responses[120]
│   │   ├── question_id: string (e.g., "Q1.1.1")
│   │   ├── selected_option: int (1-4)
│   │   ├── evidence_notes: string (optional)
│   │   └── answered_at: datetime
│   │
│   └── Results
│       ├── practice_scores[12]: float
│       ├── dimension_scores[4]: float
│       ├── overall_score: float
│       ├── weighted_scores[4]: float
│       ├── overall_weighted_score: float
│       ├── maturity_level: enum (Basic-Optimizing)
│       ├── compliance_coverage[8]: float
│       ├── strengths[]: string
│       ├── improvements[]: string
│       └── recommendations[]: string
```

7.2 Question Data Model

```
Question
├── id: string (e.g., "Q1.1.1")
├── dimension: enum (CVI, SGRM, DPE, ITR)
├── practice: string (e.g., "1.1")
├── stream: enum (A: Foundation, B: Advanced)
├── text: string
├── options[4]
│   ├── value: int (1-4)
│   ├── label: string
│   └── description: string
├── evidence_indicators: string[]
└── compliance_mapping
    ├── nist_pqc: string
    ├── cmmc: string
    ├── fedramp: string
    ├── fisma: string
    ├── nsm10: string
    ├── cnsa20: string
    ├── iso27001: string
    └── etsi_qsc: string
```

7.3 Data Storage

STORAGE TYPE	DATA	LOCATION
Client-side	Assessment responses, organization profile, theme preferences	Browser LocalStorage
File-based	Question definitions, compliance mappings, scoring thresholds	JSON/YAML in repository
Excel	Full assessment with formulas, charts, compliance tabs	`.xlsx` file (local disk)
Server-side (future)	User accounts, saved assessments, historical tracking	Database (optional)

8. External Interface Requirements

8.1 User Interfaces

See Section 5 — Frontend Requirements for detailed UI specifications.

8.2 Hardware Interfaces

Not applicable. QuantumGuard is entirely software-based.

8.3 Software Interfaces

INTERFACE	DESCRIPTION
CryptoScan CLI	Input: path to codebase; Output: SARIF report + CBOM of quantum-vulnerable algorithms
TLS Analyzer CLI	Input: hostname/IP; Output: cipher suite analysis, certificate evaluation, CNSA 2.0 compliance status
CryptoDeps CLI	Input: project dependency file (package.json, requirements.txt, etc.); Output: supply chain vulnerability report
Excel Engine	Input: 120 responses + org profile; Output: automated scores, charts, compliance mapping within workbook

8.4 Communications Interfaces

INTERFACE	DESCRIPTION
HTTPS	Web application served over TLS 1.3
GitHub API	Repository management, issues, discussions
Email	Contact form

9. Assessment Engine Specification

9.1 Question Structure

Each of the 120 questions follows this format:

- **Question text** — Clear, measurable capability question
- **4 response options** — Mapping to scores 1–4, progressing from Basic to Advanced capability
- **Evidence indicators** — Specific artifacts or demonstrations supporting the response
- **Compliance references** — Cross-references to 8 compliance frameworks

9.2 Dimension Breakdown

DIMENSION	CODE	QUESTIONS	PRACTICES
Cryptographic Visibility & Inventory	CVI	Q1–Q30	1.1, 1.2, 1.3
Strategic Governance & Risk Management	SGRM	Q31–Q60	2.1, 2.2, 2.3
Data Protection Engineering	DPE	Q61–Q90	3.1, 3.2, 3.3
Implementation & Technical Readiness	ITR	Q91–Q120	4.1, 4.2, 4.3

9.3 Scoring Algorithm (Detailed)

```
// Step 1: Practice Scores
for each practice p in [1.1, 1.2, ..., 4.3]:
    p.score = sum(p.questions[0..9].selected_option) / 10

// Step 2: Dimension Scores (Weakest Link Principle)
for each dimension d in [CVI, SGRM, DPE, ITR]:
    d.raw_score = MIN(d.practice1.score, d.practice2.score, d.practice3.score)

// Step 3: Overall Score
overall.raw_score = (CVI.raw_score + SGRM.raw_score + DPE.raw_score + ITR.raw_score)

// Step 4: Weighted Scores
profile_multiplier = calculate_profile_multiplier(org_profile)  // 0.8 - 1.5
for each dimension d:
    d.weighted_score = d.raw_score * profile_multiplier
overall.weighted_score = overall.raw_score * profile_multiplier

// Step 5: Maturity Level
function get_maturity_level(score):
    if score ≥ 4.0: return "Level 5: Optimizing"
    if score ≥ 3.5: return "Level 4: Advanced"
    if score ≥ 2.5: return "Level 3: Established"
    if score ≥ 1.5: return "Level 2: Developing"
    return "Level 1: Basic"
```

9.4 Profile Multiplier Calculation

The organization profile multiplier is derived from 5 contextual factors:

FACTOR	WEIGHT	RANGE
Industry Requirements	25%	0.8 – 1.5
Regulatory Obligations	25%	0.8 – 1.5
Organizational Scale	20%	0.8 – 1.3
Data Sensitivity	20%	0.8 – 1.5
Technology Complexity	10%	0.8 – 1.3

```
Profile Multiplier = Σ(factor_value × weight)
```

10. Compliance Mapping Module

10.1 Supported Frameworks

#	FRAMEWORK	CONTROLS MAPPED	COVERAGE SCOPE
1	NIST PQC (FIPS 203/204/205, IR 8547)	All 120 questions	Full
2	CMMC 2.0 (Levels 1–3)	All 120 questions	Full
3	FedRAMP (Low/Moderate/High)	All 120 questions	Full
4	FISMA (NIST SP 800-53 Rev. 5)	All 120 questions	Full
5	NSM-10	Selected questions	Partial
6	CNSA 2.0	Selected questions	Partial
7	ISO/IEC 27001:2022	Selected questions	Partial
8	ETSI QSC	Selected questions	Partial

10.2 Mapping Data Structure

Each question maps to specific controls in each framework. Example mapping for Q1.1.1:

```
{
  "question_id": "Q1.1.1",
  "text": "How does your organization identify cryptographic assets?",
  "mappings": {
    "nist_pqc": "NIST IR 8547 Section 3.1 'Inventory Systems'",
    "cmmc": "CM.L2-3.4.1 'Establish baseline configurations'",
    "fedramp": "CM-8 'Information System Component Inventory'",
    "fisma": "NIST SP 800-53 CM-8"
  }
}
```

11. Reporting & Visualization

11.1 Scorecard Dashboard Elements

ELEMENT	TYPE	DESCRIPTION
Overall Score	Large numeral + gauge	QuantumGuard score (1.0–4.0) with maturity level badge
Dimension Summary	4 score cards	Each showing raw score, weighted score, maturity level, %
Radar Chart	Spider/radar	4-axis comparison: org scores vs industry benchmarks
Maturity Donut	Donut chart	% of 12 practices at each maturity level
Dimension Bars	Bar chart	Side-by-side raw scores for all 4 dimensions
Practice Heatmap	4×3 grid	Color-coded maturity heatmap of all 12 practices
Strengths	Text panel	Top 3 highest-scoring practices with achievements
Improvements	Text panel	Top 3 lowest-scoring practices with recommendations
Recommendations	Dynamic text	Context-specific guidance based on scores + profile

11.2 Excel Toolkit Tabs

TAB	PURPOSE
Instructions	Getting started guide and overview
Organization Profile	Context capture and multiplier calculation
Quick Assessment	12-question rapid assessment
Dimension 1 (CVI)	30 questions for Cryptographic Visibility & Inventory
Dimension 2 (SGRM)	30 questions for Strategic Governance & Risk Management
Dimension 3 (DPE)	30 questions for Data Protection Engineering
Dimension 4 (ITR)	30 questions for Implementation & Technical Readiness
Scorecard	Automated scoring dashboard with charts
Compliance Mapping	Cross-framework mapping with coverage
Dynamic Insights	Context-specific recommendations

12. Open Source Tooling Integration

12.1 CryptoScan

FIELD	VALUE
Purpose	Discover quantum-vulnerable cryptographic algorithms in source code
Repository	CryptoScan (companion repository)
Input	File system path to codebase
Output	SARIF report, Cryptographic Bill of Materials (CBOM)
QuantumGuard Dimension	Dimension 1: CVI — Practice 1.1 (Cryptographic Discovery & Inventory Management)
Detectable Algorithms	RSA, ECDSA, ECDH, DSA, DH, SHA-1, MD5, 3DES, RC4, and other quantum-vulnerable primitives

12.2 TLS Analyzer

FIELD	VALUE
Purpose	Evaluate TLS/SSL configurations and certificates for quantum vulnerability
Repository	TLS Analyzer (companion repository)
Input	Hostname/IP address
Output	Cipher suite analysis, certificate evaluation, CNSA 2.0 compliance status
QuantumGuard Dimension	Dimension 3: DPE — Practice 3.3 (Transit Security & Protocol Management)

12.3 CryptoDeps

FIELD	VALUE
Purpose	Identify quantum-vulnerable algorithms in software supply chain dependencies
Repository	CryptoDeps (companion repository)
Input	Dependency manifest files (package.json, requirements.txt, pom.xml, etc.)
Output	Supply chain vulnerability report with affected packages and algorithms
QuantumGuard Dimension	Dimension 1: CVI — Practice 1.3 (Cryptographic Dependency Mapping)

13. Security Requirements

13.1 Data Protection

ID	REQUIREMENT
SR-001	Assessment data stored client-side shall be encrypted or obfuscated in LocalStorage
SR-002	No assessment data shall be transmitted to external servers without explicit user consent
SR-003	Web application shall be served exclusively over HTTPS (TLS 1.2+)
SR-004	Exported PDF/Excel reports shall not contain metadata exposing internal system information

13.2 Application Security

ID	REQUIREMENT
SR-005	Web application shall implement Content Security Policy (CSP) headers
SR-006	All user inputs shall be sanitized to prevent XSS and injection attacks
SR-007	No third-party analytics or tracking shall be implemented without disclosure
SR-008	Dependencies shall be regularly audited for known vulnerabilities

13.3 Repository Security

ID	REQUIREMENT
SR-009	No secrets, credentials, or API keys shall be committed to the repository
SR-010	Contribution guidelines shall include security review requirements
SR-011	Releases shall be signed/tagged for integrity verification

14. Appendices

Appendix A: QuantumGuard Dimension Summary

DIM	NAME	CODE	FOCUS	PRACTICES
1	Cryptographic Visibility & Inventory	CVI	Understanding and cataloging cryptographic assets	1.1 Discovery & Inventory, 1.2 Vulnerability Assessment, 1.3 Dependency Mapping
2	Strategic Governance & Risk Management	SGRM	Leadership commitment and systematic risk management	2.1 Executive Leadership & Policy, 2.2 Risk Assessment & Compliance, 2.3 Third-Party & Supply Chain
3	Data Protection Engineering	DPE	Technical implementation of quantum-safe protections	3.1 Data Classification & Protection, 3.2 Storage Security & Encryption, 3.3 Transit Security & Protocols
4	Implementation & Technical Readiness	ITR	Execution capabilities for quantum-safe deployment	4.1 Infrastructure Assessment, 4.2 Implementation Capability, 4.3 Testing & Validation

Appendix B: Maturity Level Summary

LEVEL	NAME	SCORE RANGE	KEY CHARACTERISTICS
1	Basic	1.0 – 1.4	Ad-hoc, limited awareness, reactive
2	Developing	1.5 – 2.4	Initial structure, formal recognition, growing investment
3	Established	2.5 – 3.4	Systematic, comprehensive, cross-functional, sustained
4	Advanced	3.5 – 3.9	Optimized, predictive, proactive, industry-leading
5	Optimizing	4.0	Innovation, thought leadership, strategic advantage

Appendix C: Implementation Phases

PHASE	DURATION	ACTIVITIES
Discovery & Assessment	3–6 months	Complete QuantumGuard assessment, identify vulnerabilities, establish baselines
Planning & Prioritization	2–4 months	Create strategy, prioritize systems, allocate resources, secure approval
Pilot Implementation	6–12 months	Deploy in test environments, validate, refine processes, build capabilities
Enterprise Rollout	12–24 months	Systematic deployment, continuous monitoring, regular reassessment

Appendix D: Typical Maturity Advancement Timelines

TRANSITION	DURATION	FOCUS
Level 1 → 2	12–18 months	Structured processes, formal governance
Level 2 → 3	18–24 months	Systematic implementation, comprehensive coverage
Level 3 → 4	24–36 months	Advanced optimization, predictive capabilities
Level 4 → 5	36+ months	Industry leadership, breakthrough innovation

Appendix E: Glossary of Quantum Cryptography Terms

TERM	DEFINITION
Post-Quantum Cryptography (PQC)	Cryptographic algorithms believed to be secure against both classical and quantum computers
Harvest Now, Decrypt Later	Attack strategy where encrypted data is collected today for future quantum decryption
Cryptographic Agility	Ability to quickly switch between cryptographic algorithms without major system changes
Hybrid Cryptography	Using both classical and post-quantum algorithms together for defense-in-depth
CRQC	Cryptographically-Relevant Quantum Computer — a quantum computer powerful enough to break current encryption
Lattice-Based Cryptography	PQC approach based on mathematical lattice problems (basis for ML-KEM, ML-DSA)
Hash-Based Signatures	PQC approach using hash functions for digital signatures (basis for SLH-DSA)

Document Control

VERSION	DATE	AUTHOR	CHANGES
1.0	2026-03-15	QuantumGuard Documentation Team	Initial SRS creation